Six Forbidden Prime Classes in an Exponential Sequence

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Abstract

We prove the six excluded prime residue classes conjectured in OEIS A237930. Each excluded prime makes three a quadratic residue and seven a quadratic nonresidue, which prevents it from dividing any term $(7\cdot 3^n-1)/2$. We also give the exact finite multiplicative-order criterion for every other odd prime.

1 The six forbidden classes

Write

$$A(n) = 3^{n+1} + \frac{3^n-1}{2} = \frac{7\cdot 3^n - 1}{2}, \qquad n \geq 0.$$

The April 13, 2020 comment in OEIS A237930 conjectures that no prime congruent to $11, 13, 23, 61, 71$, or 73 modulo 84 divides any term. The listed classes are interpreted as six integers; the missing comma between 61 and 71 in the entry is typographical.

For an odd prime p and integer a coprime to p, denote the Legendre symbol by $\left(\frac{a}{p}\right)$.

Theorem 1. *For every $n \geq 0$, no prime divisor of $A(n)$ is congruent to any of*

$$11, 13, 23, 61, 71, 73 \pmod{84}.$$

Proof. Quadratic reciprocity and its supplementary laws give

$$\left(\frac{3}{p}\right) = \begin{cases} 1, & p \equiv 1, 11 \pmod{12}, \\ -1, & p \equiv 5, 7 \pmod{12}, \end{cases} \qquad \left(\frac{7}{p}\right) = (-1)^{(p-1)/2}\left(\frac{p}{7}\right).$$

The nonzero squares modulo seven are $1, 2, 4$. Applying these rules to the unit classes modulo 84 gives

$$\left(\frac{3}{p}\right) = 1, \quad \left(\frac{7}{p}\right) = -1 \quad \Longleftrightarrow \quad p \equiv 11, 13, 23, 61, 71, 73 \pmod{84}.$$

For transparency, the associated local residues are

$p \bmod 84$	11	13	23	61	71	73
$p \bmod 12$	11	1	11	1	11	1
$p \bmod 7$	4	6	2	5	1	3
$p \bmod 4$	3	1	3	1	3	1

If such a p divided $A(n)$, then $7 \cdot 3^n \equiv 1 \pmod{p}$. Taking Legendre symbols would give

$$1 = \left(\frac{7}{p}\right)\left(\frac{3}{p}\right)^n = -1,$$

which is impossible. □

2 The exact incidence condition for other primes

The obstruction proves precisely the six-class conjecture; it need not characterize all primes that never divide the sequence. An exact finite test for each remaining prime is as follows. The prime 2 divides $A(n)$ exactly when n is odd, since $7 \cdot 3^n - 1 \equiv 0 \pmod 4$ exactly then. The prime 3 divides only $A(0) = 3$, and 7 divides no term.

For an odd prime $p \neq 3, 7$, put $t = \operatorname{ord}_p(3)$. If the finite set

$$H_p = \{0 \leq h < t : 3^h \equiv 7^{-1} \pmod p\}$$

contains an element h, it is unique, and

$$p \mid A(n) \iff n \equiv h \pmod t.$$

If H_p is empty, p divides no term. This follows immediately from the periodic and distinct powers $3^0, \ldots, 3^{t-1}$. Thus the sequence's individual prime-divisibility questions are also completely decidable by finite arithmetic.

3 Independent checks

The accompanying verifier first classifies all unit residues modulo 84, then checks the two quadratic symbols by the independent Euler criterion for all relevant primes through 200000. This includes 4513 primes in the excluded classes. For every excluded prime below 5000, it independently traverses the entire orbit of powers of three and checks absence of a solution; these cycles contain 134995 residues in total. All 1001 official sequence terms agree with the exponential formula and the independent affine recurrence $A(0) = 3$, $A(n+1) = 3A(n) + 1$. The all-index exclusion is proved above, not inferred from those finite tests.

References

[1] The OEIS Foundation Inc., A237930 and its b-file, accessed October 26, 2026.